COMSATS University Islamabad, Lahore Campus
Department of Computer Engineering
1.5 KM Defence Road, Off Raiwind Road, Lahore

Version: v.4.0

Course Descriptive File

Course Code and Title	CPE325-Control Systems		
Credit Hours	4 (3,1)	**Semester**	7th
Prerequisites	CPE223	**Co requisites**	N/A
WK			
Course Type			

1 Course Outline as per SoS

Introduction, examples of everyday control systems, definition and classification, open loop and closed loop control systems. Laplace Transformation- Definition, Laplace Transformation of common signals in time domain, Inverse Laplace Transform, Laplace Theorems, Examples and solution of ordinary differential equations employing Laplace transforms. Identification of systems, Mathematical modeling by physical laws, Modeling of Mechanical systems, Electrical and Electronic systems, Electromechanical systems, Hydraulic systems, and dynamic systems. Block diagrams reduction Techniques, Signal flow graphs- Mason's gain formula. Impulse and Step response along with Time response 1st and 2nd order systems, Calculation of overshoot, undershoot, time constant, rise time, peak time, settling time etc. Steady state errors in Control Systems, positional, velocity and acceleration error constants, Routh-Hurwitz criterion for stability. PID controllers, effect on the performance of control systems by individual elements of PID controllers. Introduction to design using root locus method and analysis, Frequency domain analysis, Bode analysis, Technique of drawing Bode plot. Phase margin, Gain margins and Design using Bode plots. Introduction to Nyquist analysis, complex analysis, Encirclement and enclosure of poles and zeros, Principle of Argument, Stability analysis, State space, Introduction to state space analysis of systems presented by Transfer functions. Controllability and Observability, and Solution of Time invariant state equations.

2 Course Objectives as per SoS

This course familiarizes students with the basic concepts of control systems. This module is intended to lay a foundation for designing advanced control systems. The students in this course will learn time-domain and frequency-domain methods for the analysis, modeling and design of continuous-time control systems.

3 Suggested Books

Textbook

1. Modern Control Engineering Latest Edition by Katsuhiko Ogata.

Reference Books

2. Control System Engineering Latest Edition by Norman S. Nise
3. Modern Control Systems Latest Edition by Richard C. Dorf and Robert H. Bishop

4	Course Learning Outcomes (CLOs)

After successful completion of this module, you will be able to:

Theory CLOs:

1. Analyze the behavior of open-loop and closed-loop control systems using time-domain and frequency-domain techniques. (C4-PLO2)
2. Design PID controllers for feedback systems to meet specific performance criteria, such as stability, response time, and accuracy. (C5-PLO3)
3. Evaluate the stability and performance of control systems using root locus, Bode plot, and Nyquist plot techniques. (C5-PLO1)
4. Conduct an experiment or simulation to model a control system for a real-life engineering problem, considering safety, environmental, and societal factors. (C6-PLO7)

Lab CLOs:

5. Design, simulate and implement control systems using modern tools like MATLAB/Simulink, performing stability and performance analysis. (PLO5, P5)

6. Prepare and present technical lab reports effectively, demonstrating clear communication of control system experiment results. (PLO10, A3)

7. Demonstrate a commitment to addressing complex engineering challenges by valuing continuous learning and the adaptability required to engage with emerging control system technologies. (PLO12, A3)

CLO	PLO	Bloom Taxonomy Domain	Bloom Taxonomy Level
1	PLO2– Problem Analysis	Cognitive	C4 – Analysis
2	PLO3– Design/Development of Solutions	Cognitive	C5 – Synthesis
3	PLO1– Engineering Knowledge	Cognitive	C5 – Synthesis
4	PLO7– Environment and Sustainability	Cognitive	C6 – Evaluation
5	PLO5– Modern Tool Usage	Psychomoto r	P5 – Complex response
6	PLO10– Communication	Affective	A3 - Valuing
7	PLO12– Lifelong Learning	Affective	A3 - Valuing

PLOs Coverage Explanation

PLO2 - Problem Analysis This is relevant for analyzing control systems, where students must break down complex system behaviors and solve issues related to stability and performance.
PLO3 - Design/Development of Solutions Designing control systems (e.g., PID controllers) is a core aspect of control systems engineering, making this PLO central to the course.
PLO1 - Engineering Knowledge The course involves applying core engineering and mathematical concepts in control systems analysis, which is directly aligned with PLO1.
PLO7 - Environment and Sustainability Control systems are often used in contexts that have significant environmental and safety implications, making it necessary to consider these factors in design and simulation, thus connecting with PLO7.
PLO5 - Modern Tool Usage Control systems heavily rely on modern engineering tools like MATLAB and Simulink for simulation and implementation, making PLO5 critical for lab outcomes.
PLO10 – Communication Writing lab reports and communicating technical details effectively are essential skills for engineers, making PLO10 vital for the lab component.
PLO12 - Lifelong Learning The rationale for adding this affective domain CLO focused on complex engineering problems and lifelong learning in the Control Systems course is rooted in preparing students to adapt to rapid technological advancements in engineering fields.

5	**Marks Breakup**

Theory

Quizzes	15%
Assignments	10%
Midterm Exam	25%
Terminal Exam	50%
Total (theory)	**100 %**

Lab

Lab Assignments: i. Lab Assignment 1 Marks (Lab marks from Labs 1-3) ii. Lab Assignment 2 Marks (Lab marks from Labs 4-6) iii. Lab Assignment 3 Marks (Lab marks from Labs 7-9) iv. Lab Assignment 4 Marks (Lab marks from Labs 10-12)		25%
Lab Mid Term = 0.5 × (Lab Mid Term exam) + 0.5× (average of lab evaluation of Lab 1-6)		25%
Lab Terminal		50%
☐ **CEA**	0.5*(Complex Engineering Problem) +0.375*(average of lab evaluation of Lab 7-12) + 0.125*(average of lab evaluation of Lab 1-6)	
☐ **OEL**	0.1*(Open Ended Lab) +0.4*(Lab Terminal Exam) +0.375 *(average of lab evaluation of Lab 7-12) + 0.125*(average of lab evaluation of Lab 1-6)	
☐ **Project Based**	0.2*(Lab Project) +0.3*(Lab Terminal Exam) +0.375*(average of lab evaluation of Lab 7-12) + 0.125*(average of lab evaluation of Lab 1-6)	
☐ **Conventional**	0.5*(Lab Terminal Exam) +0.375*(average of lab evaluation of Lab 7-12) + 0.125*(average of lab evaluation of Lab 1-6)	
	Total (lab)	**100%**

Final marks	**Theory marks × 0.75 + Lab marks ×0.25**

Version No.	v.4.0
Revision Date	Spring 2025
Last Modification Date	Fall 2023
Modification Details	Updating of CLO statements, Bloom taxonomy levels and PLO mapping
Recommended by	Dr. Mujtaba Hussain Jaffery
Approved by	DQEC

Annexure – I: Lecture Plan

Lectures	Topic	CLO	Contact Hours	Students Learning Hours	Assessment	Bloom Taxonomy
1	Some fundamental terms, examples of Control Systems	CLO 1	3	3	Q1, Q2 Midterm, Terminal	C4,C5
2-3	Laplace transformation, Partial fraction, Solving LTI differential equations, Transfer Function		3	3		
4-6	Mathematical modelling of electrical and electronic systems, mechanical systems, electromechanical systems, hydraulic systems, and dynamic systems		9	9		
7-8	Block Diagram, Block Diagram Reduction Algebra	CLO 2	6	6		
9	Signal Flow Graphs (Mason's gain Formula)		3	3		
10-12	Transient response analysis of first order and second order systems					
13	Steady State Error Analysis for feedback systems					
14-15	Routh-Hurwitz Stability Criteria		6	6		
16-19	Design Via Root Locus, Minimizing Steady State Error and Improving Transient Response Via Cascade Compensation (Lead and Lag compensation)		6	6	Q3,Q4 PBL, Terminal	C5& C6
20-23	Introduction to PID Controllers, Tuning Rules for PID Controller, PID control for VTOL, HVAC and other related systems.		6	6		

24-25	Bode plots and linear systems frequency response	CLO 3	6	6		
26-27	Lag-Lead Compensation	CLO 4	6	6		
28-29	Nyquist stability criteria and stability margins.		6	6		
30-31	State Space Approach for control systems, Controller design using state feedback approach e.g. Pole placement controller		6	6		
32	Introduction to Digital Control Systems		1.5	1.5		

Annexure – I: List of Experiments

Sr. No.	Topics Covered	CLO	Bloom Taxonomy
1	Mathematical Modelling of Mechanical, Electrical, Electronic and Electromechanical Systems using MATLAB/LabVIEW	CLO5	P5
2	Block Diagram Reduction Techniques and Measuring Time-Domain Performance Parameters & Effect of Disturbance in Open/Closed-Loop Control System using MATLAB	CLO5	P5
3	Design of Proportional-Integral-Derivative (PID) Controller using MATLAB	CLO5	P5
4	Introduction to Vertical Take-off and landing (VTOL) and Design of a Current Controller using LabVIEW	CLO5	P5
5	Mathematical Modelling of the Vertical Take-Off and Landing System for the flight Control	CLO5	P5

	Operation and its validation using LabVIEW		
6	Design of a PID Controller for Pitch Control operation of VTOL	CLO5	P5
7	Design of a Switch (ON/OFF) Controller and Mathematical Modelling of QNET HVAC System and its validation using LabVIEW	CLO5	P5
8	Mathematical Modelling of the QNET HVAC System and Design of PI Controller for the System	CLO5	P5
9	Design of Proportional Controller and Lead/Lag Compensator using Root Locus Method on MATLAB	CLO5	P5
10	State Feedback Controller Design for a DC Motor Using Pole Placement Technique	CLO5	P5
11	Introduction to Rotary Pendulum System and to analyse the dynamical behaviour of the System using LabVIEW	CLO5	P5
12	Design and Implementation of a Hybrid Controller for Rotary Pendulum	CLO5	P5
	PLO10 – Communication Writing lab reports and communicating technical details effectively are essential skills for engineers, making PLO10 vital for the lab component. **PLO12 - Lifelong Learning** The rationale for adding this affective domain CLO focused on complex engineering problems and lifelong learning in the Control Systems course is rooted in preparing students to adapt to rapid technological advancements in engineering fields.	CLO6+CLO7	A3